ASSIGNMENT-1

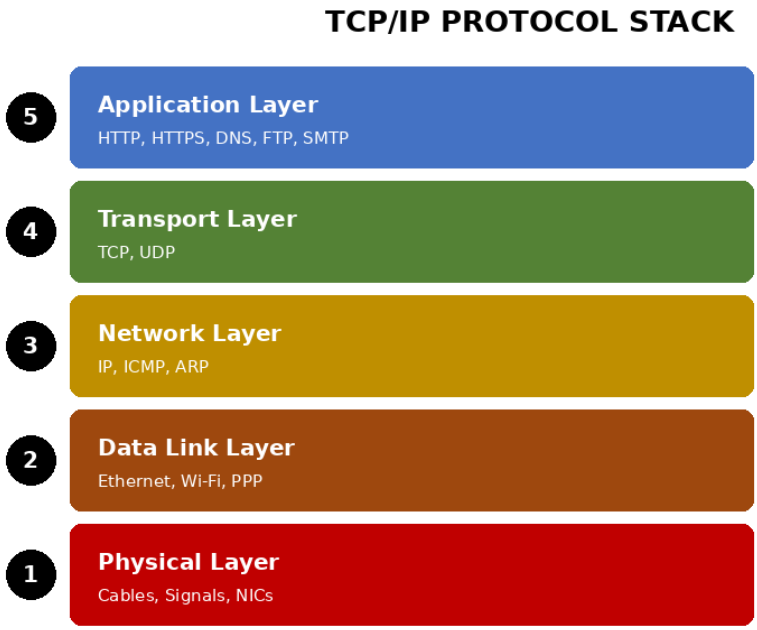
PCCST501 – Computer Networks

How the Internet Delivers a Web Page

Website chosen: https://www.amazon.com

1. Draw the TCP/IP protocol stack and explain the role of each layer

The TCP/IP protocol suite is organised as five layers, each providing a specific service to the layer above it and using the services of the layer below it (this idea is called protocol layering). When a user opens www.amazon.com, all five layers take part in getting the page from Amazon's server to the browser.



Layer	Key Protocols	Main Responsibility
Application	HTTP, HTTPS, DNS, FTP, SMTP	Provides process-to-process communication; generates the request the user actually cares about, e.g. the HTTP GET for the Amazon home page.
Transport	TCP, UDP	Provides end-to-end delivery between the two hosts; TCP adds reliability, ordering, flow and congestion control.

Network	IP, ICMP, ARP	Provides host-to-host addressing and routing; moves the datagram across multiple networks toward the destination.
Data Link	Ethernet, Wi-Fi, PPP	Provides node-to-node (hop-to-hop) delivery on a single link; frames the datagram and adds MAC addressing.
Physical	Cables, radio signals, connectors	Converts the frame's bits into electrical, optical or radio signals for transmission over the medium.

Two concepts tie the layers together. Encapsulation means that as data moves down the stack at the source host, each layer wraps the data from the layer above with its own header (the message becomes a segment, then a datagram, then a frame). Decapsulation is the reverse process performed at the destination. A router in the path only decapsulates up to the network layer, chooses the next link, and re-encapsulates the datagram in a new frame, it does not need the transport or application layers at all.

2. Explain the role of DNS in converting the domain name into an IP Address

Computers route traffic using IP addresses, not names, so when a user type www.amazon.com the browser must first translate that name into an IP address. This is the job of the Domain Name System (DNS), a distributed, hierarchical client-server directory service.

How the lookup happens: the browser calls a DNS client called a resolver. The resolver first checks the browser cache and the operating-system cache; if the mapping is not already known, it queries a local (recursive) DNS server, usually run by the ISP.

- If the local server already has the mapping cached, it replies immediately.
- If not, the local server queries a root server, which points it to the .com top-level-domain (TLD) server.
- The TLD server points it to the authoritative name server for amazon.com.
- The authoritative server returns the actual IP address, which is relayed back through the chain to the browser.

Every answer is returned with a Time to Live (TTL) so that it can be cached for a limited period, which speeds up future lookups and reduces load on the DNS hierarchy. Without DNS, users would have to memorise numeric addresses for every website they visit.

3. Describe how HTTP is used to request and receive the webpage

Once the browser has Amazon's IP address, it uses the Hypertext Transfer Protocol (HTTP) to fetch the page. HTTP is a client-server, request-response protocol that relies

on TCP for delivery, so a TCP connection must exist before any HTTP message can be exchanged.

- The browser opens a TCP connection to the server, normally on port 80 (or 443 for HTTPS, where a TLS handshake also happens).
- The browser sends a request message consisting of a request line, header lines, a blank line, and (for GET) an empty body, for example: GET / HTTP/1.1 Host: www.amazon.com
- The server processes the request and locates or generates the requested resource.
- The server replies with a response message: a status line (e.g. HTTP/1.1 200 OK), header lines describing the content, and a body containing the HTML document.
- The browser then issues further GET requests for embedded objects referenced in the HTML, CSS files, JavaScript, and images and renders the complete page once these arrive.

HTTP/1.1 uses persistent connections by default, so several of these object requests can reuse the same TCP connection instead of opening a new one each time, which reduces overhead considerably on an image-heavy page like an Amazon product listing.

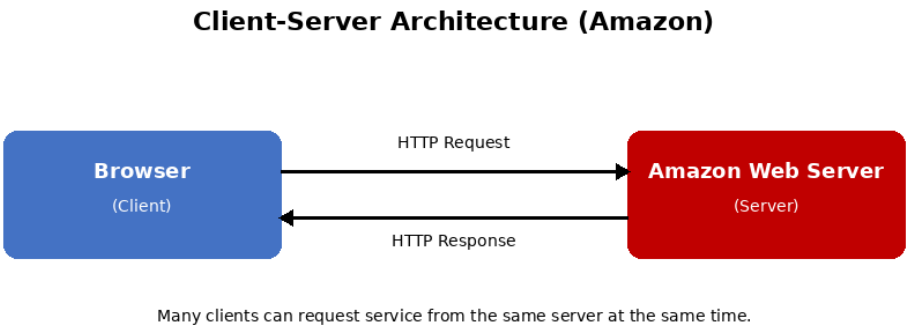
4. Explain how TCP provides reliable communication for HTTP

HTTP itself has no mechanism for detecting lost or corrupted data, it simply trusts the transport layer to deliver its messages correctly. That reliability is supplied by TCP, a connection-oriented, byte-stream protocol with the following features:

- Connection establishment (three-way handshake) : SYN, SYN-ACK, ACK synchronises sequence numbers before any HTTP data is sent.
- Error control : every byte is numbered and acknowledged; corrupted or missing segments are detected and retransmitted.
- Ordered delivery : sequence numbers let the receiving TCP reassemble segments in the correct order even if they arrive out of sequence.
- Flow control : a sliding-window mechanism prevents the server from overwhelming the browser's receive buffer.
- Congestion control : TCP throttles its sending rate when it detects network congestion, protecting shared links.

Because of these guarantees, the HTML, CSS, JavaScript and images that make up the Amazon page always arrive complete and in the correct order, even though the underlying network layer (IP) offers no such guarantee on its own.

5. Identify whether the application follows a Client–Server or Peer-to-Peer Architecture and justify your answer.



Architecture used: Client–Server.

In the client-server paradigm, a server process runs continuously and waits for connections, while a client process starts only when the user needs the service, sends a request, and stops once it has been served. This matches Amazon exactly:

- Amazon's web servers are always running and listening for connections, they never initiate contact with a browser on their own.
- The browser (client) is started by the user, sends a request, receives the page, and can be closed afterwards.
- The relationship is asymmetric: the browser cannot act as a server for another browser, and Amazon's server does not behave like a client toward the user.
- Amazon can serve millions of clients concurrently from a farm of powerful servers, which is the typical (and typical drawback of) client-server design the server side must be resourced to handle the concentrated load.

This is different from a peer-to-peer (P2P) system such as BitTorrent, where every node can act as both a provider and a consumer of the same resource and no single machine needs to be running permanently.

6. Compare HTTP and FTP with at least four differences

Both HTTP and FTP are standard, client-server application-layer protocols that run over TCP, but they were designed for different jobs.

Point of comparison	HTTP	FTP
Purpose	Requests and delivers web pages and their embedded objects (HTML, CSS, JS, images).	Transfers whole files between a client and a server, including uploading and downloading.
Connections used	Normally one TCP connection is used to carry both the request and the	Uses two separate TCP connections at the same time: a control connection (port 21) for commands and a data

	response (persistent by default in HTTP/1.1).	connection (port 20) for the actual file.
State	Stateless by design — each request/response pair is independent; cookies are used to fake state when needed.	Stateful for the duration of the session — the control connection stays open and remembers the current directory, login status, etc.
Well-known port	80 for HTTP, 443 for HTTPS.	21 for control, 20 for data.
Data handling	Sends the object as a single response body; the browser interprets its content type from a header.	Let's the client specify file type (ASCII, EBCDIC, image), data structure and transmission mode before the transfer.
Authentication	Not required to fetch a public page (login is only needed for personalised content).	Traditionally requires a USER/PASS login before any file operation is allowed.

7. Explain how cookies improve user experience with one real-world example

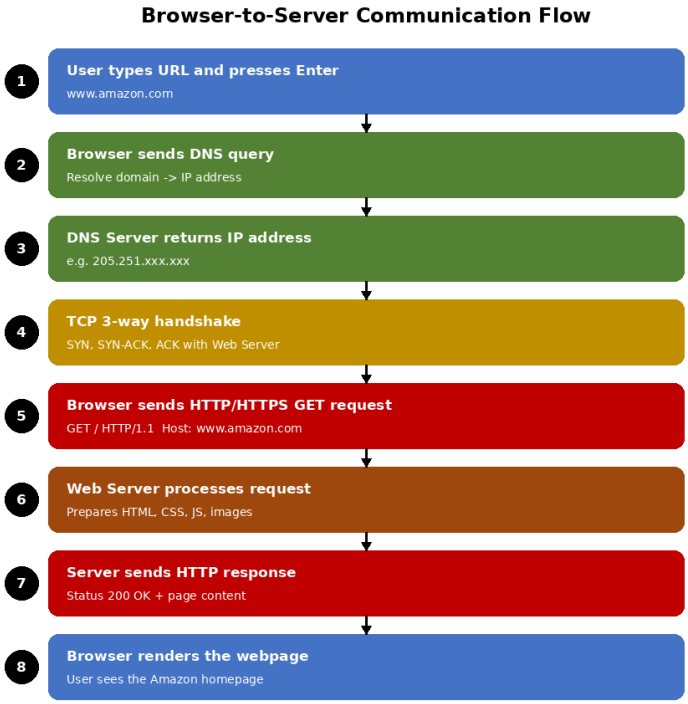
HTTP was originally designed to be stateless; the server treats every request as if it came from a brand-new client. A cookie is a small piece of data that the server asks the browser to store and to send back with every later request, which effectively lets an otherwise stateless protocol remember information about a specific user.

Creating and using a cookie: when the server first replies to a client, it can include a Set-Cookie header containing an identifier. The browser stores this value and automatically attaches it, via a Cookie header, to every subsequent request to that same domain. The server uses the identifier to look up whatever information it has stored about that visitor.

Real-world example: Amazon shopping cart

When a shopper adds a product to the cart on Amazon, the server creates (or updates) a cart record and gives the browser a cookie holding the cart's identifier. As the shopper selects more items, each request carries that cookie back to the server, which updates the same cart. When the shopper checks out, the server retrieves the whole cart using the cookie value, without the shopper having to re-enter each item. On a later visit, the same mechanism lets Amazon keep the shopper signed in, recommend products based on browsing history, and remember preferences such as language or delivery address ,all of which make the site faster and more personal to use.

8. Draw a flow diagram showing the complete communication from the browser to the web server and back



9. **Write a brief conclusion (100–150 words) describing the importance of application layer protocols in everyday Internet usage.**

Application-layer protocols are what make the Internet useful to ordinary people, even though they depend entirely on the layers beneath them to function. In the case of Amazon, DNS quietly converts a memorable domain name into a routable IP address, TCP builds a reliable channel across an unreliable network, and HTTP uses that channel to request and receive the actual page content, with cookies adding the personalised, stateful experience that modern e-commerce relies on. Every one of these steps happens within a few hundred milliseconds and is completely invisible to the user, who only sees the finished web page appear. Understanding this chain ,from typing a URL to seeing a rendered page ,shows why application-layer protocols, working together with the rest of the TCP/IP stack, are essential to everyday Internet usage, whether it is browsing a store, sending an email, or transferring a file.